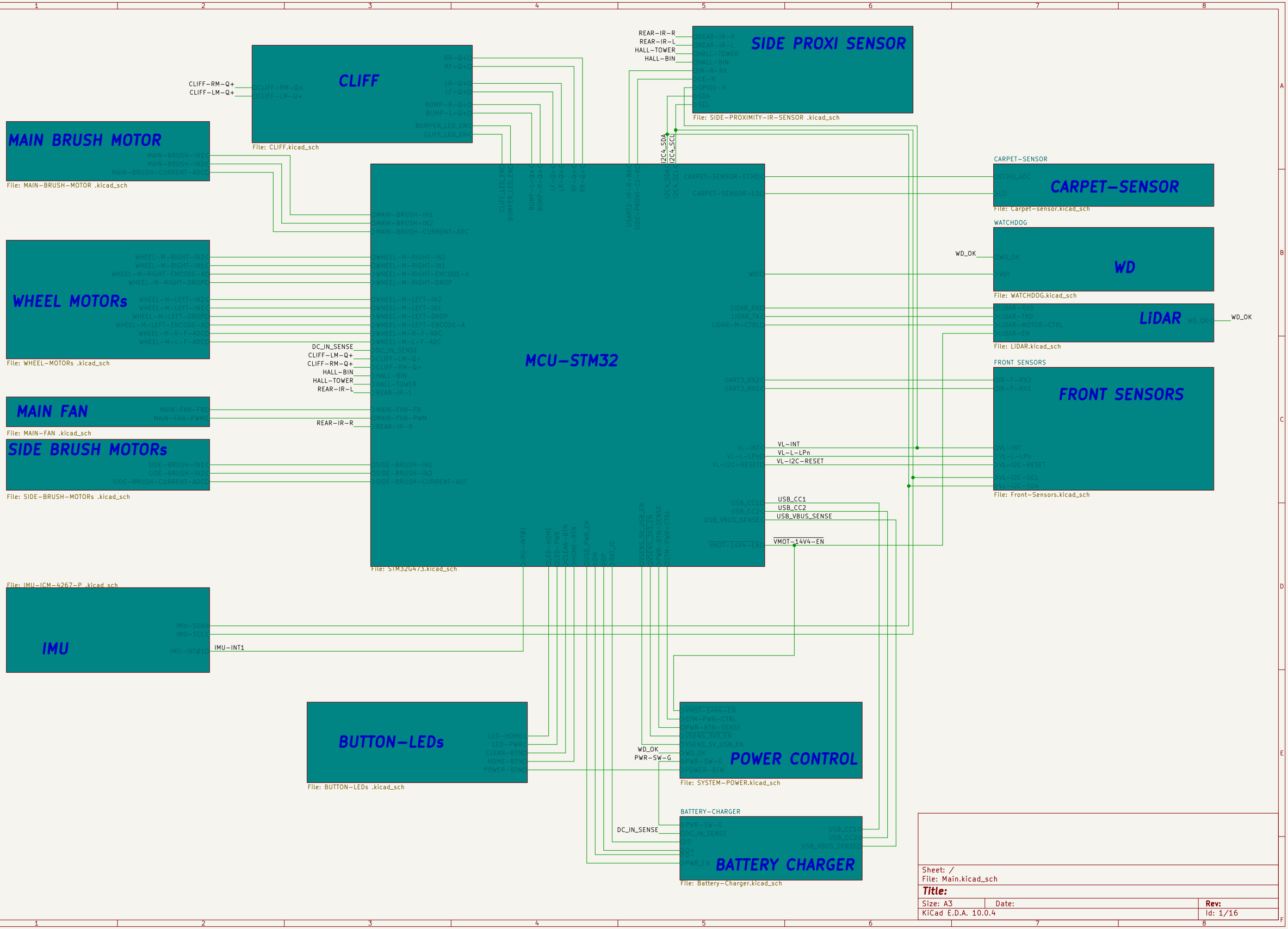
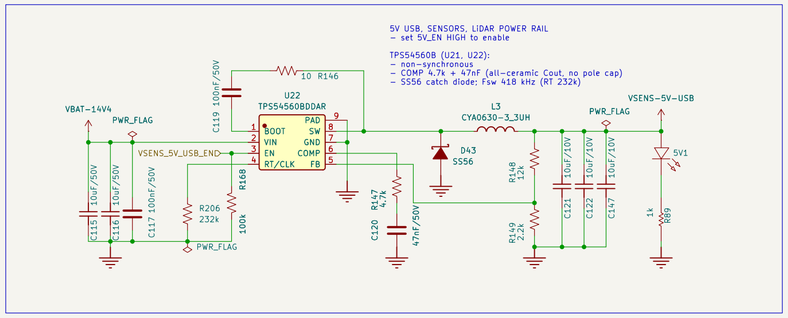
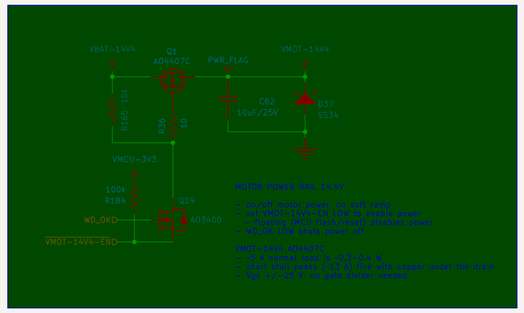
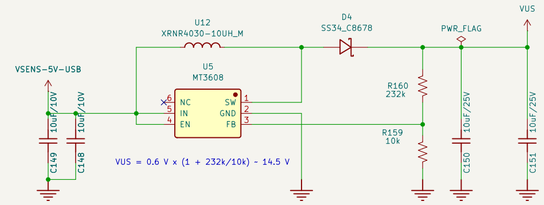
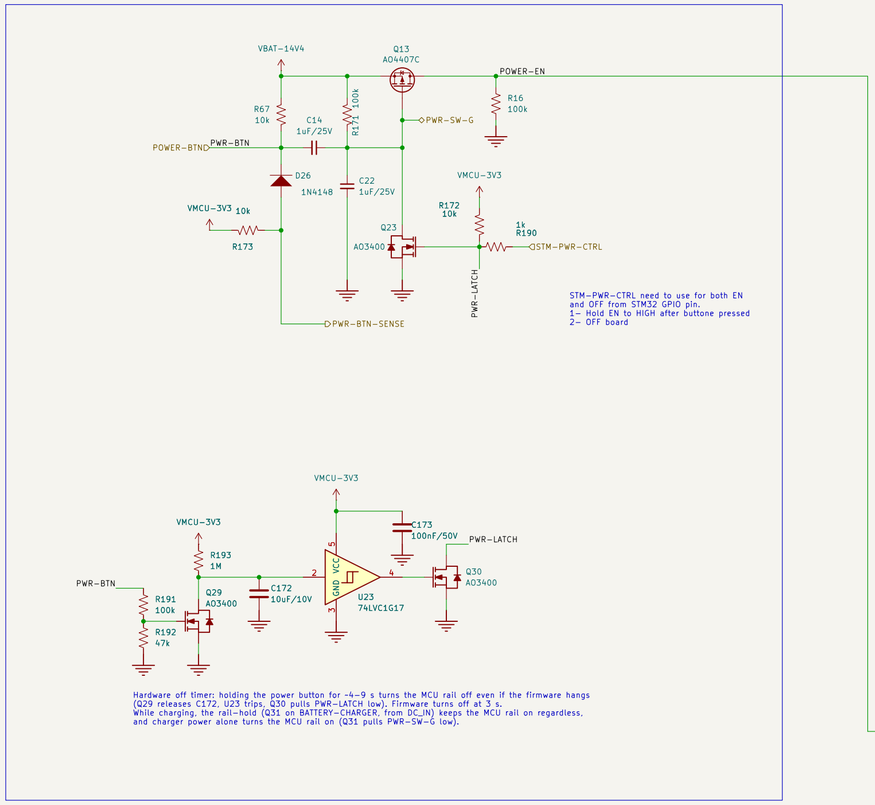
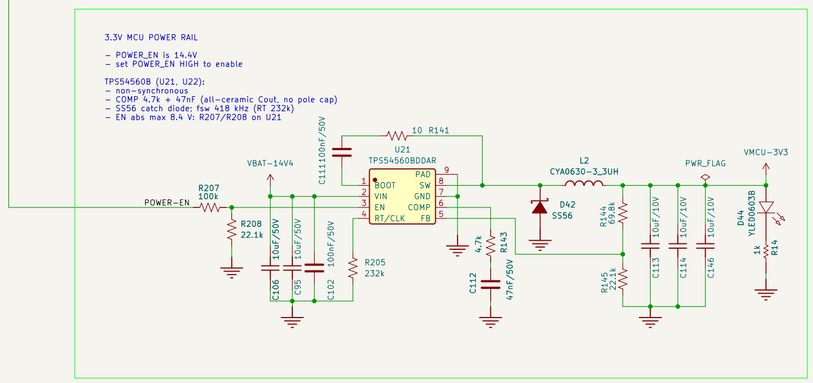

CLIFF
File: CLIFF.kicad_sch
SIDE PROXI SENSOR
File: SIDE-PROXIMITY-IR-SENSOR .kicad_sch
MAIN BRUSH MOTOR
File: MAIN-BRUSH-MOTOR .kicad_sch
MCU-STM32
File: STM32G473.kicad_sch
CARPET-SENSOR
File: Carpet-sensor.kicad_sch
WATCHDOG
WD
File: WATCHDOG.kicad_sch
WHEEL MOTORs
File: WHEEL-MOTORs .kicad_sch
LiDAR
File: LiDAR.kicad_sch
FRONT SENSORS
File: Front-Sensors.kicad_sch
MAIN FAN
File: MAIN-FAN .kicad_sch
SIDE BRUSH MOTORs
File: SIDE-BRUSH-MOTORs .kicad_sch
IMU
File: IMU-ICM-4267-P .kicad_sch
BUTTON-LEDs
File: BUTTON-LEDs .kicad_sch
POWER CONTROL
File: SYSTEM-POWER.kicad_sch
BATTERY CHARGER
File: Battery-Charger.kicad_sch
Sheet: /
File: Main.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/16

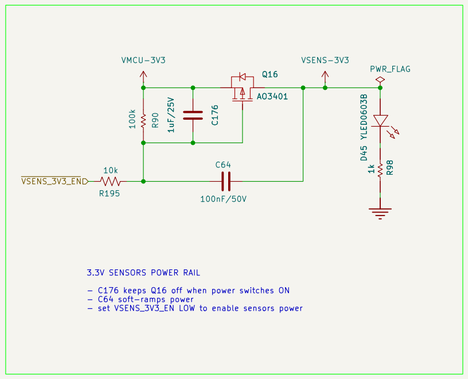


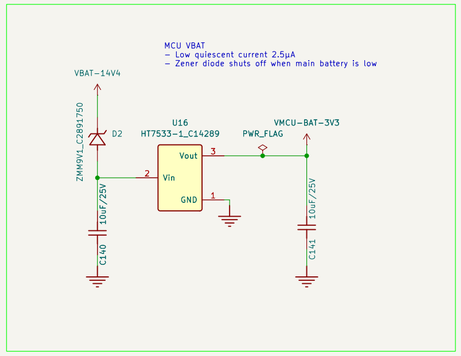












PCB mounting standoffs (Wurth WA-SMSI, M3), moved here from the removed M.2 sheet.

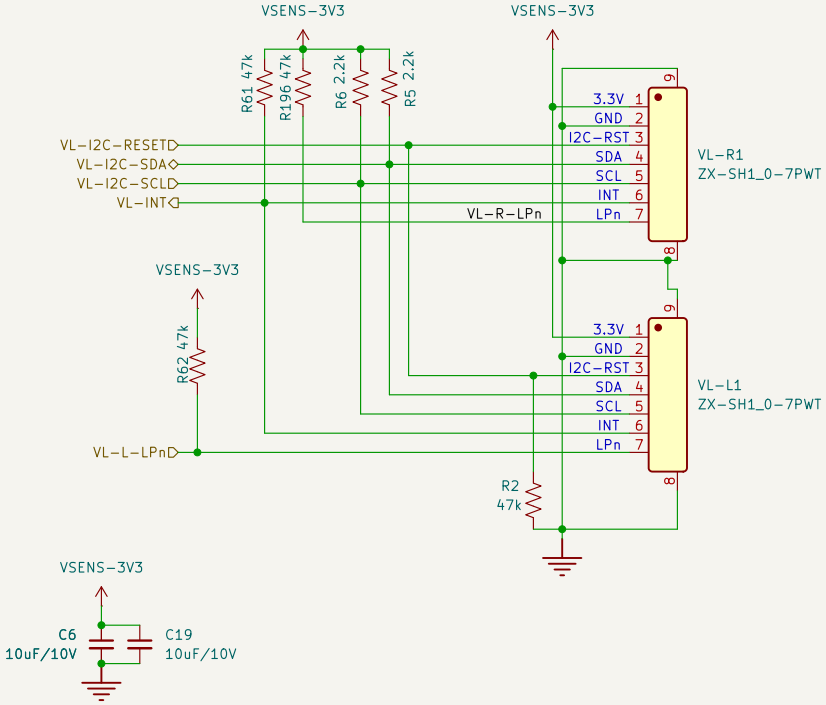
MP1 MP2 MP3 MP4 MP5 MP6

Sheet: /POWER/
File: SYSTEM-POWER.kicad_sch

Title:

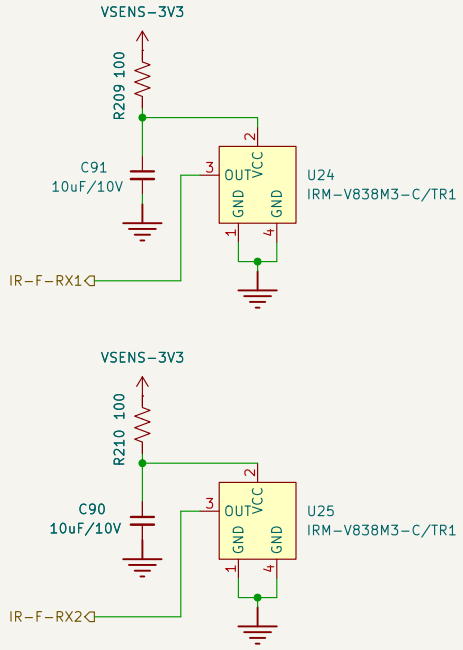
Size: A2 | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 2/16

2x VL53L7 ToF BOARDS



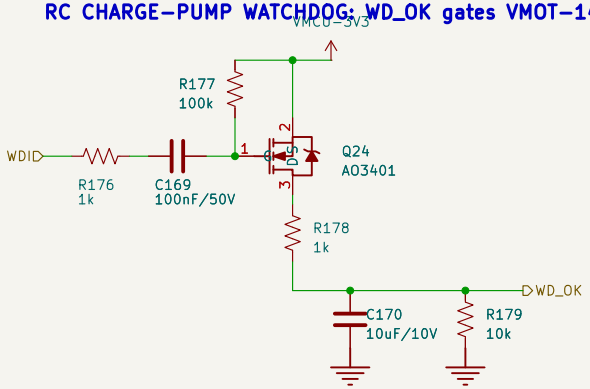
2 IR RECEIVERS

U24 left, U25 right (robot view). Side-looking: put both at the board front edge, lenses facing forward, ~10 mm apart, with a baffle between them in the bumper window. – 90 deg FoV



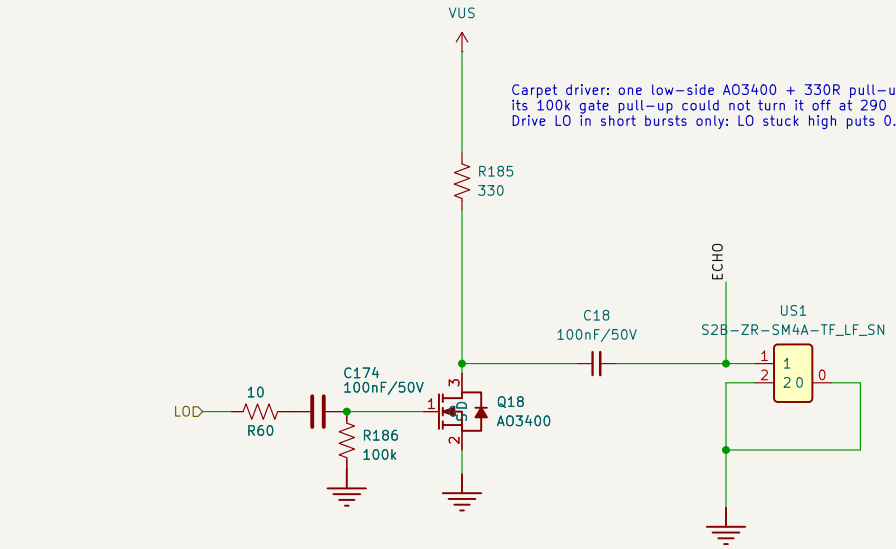
Sheet: /FRONT SENSORS/	
File: Front-Sensors.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 4/16

RC CHARGE-PUMP WATCHDOG: WD_OK gates VMOT-14V4, VM-5V-LIDAR and VM-5V-WATER-PUMP



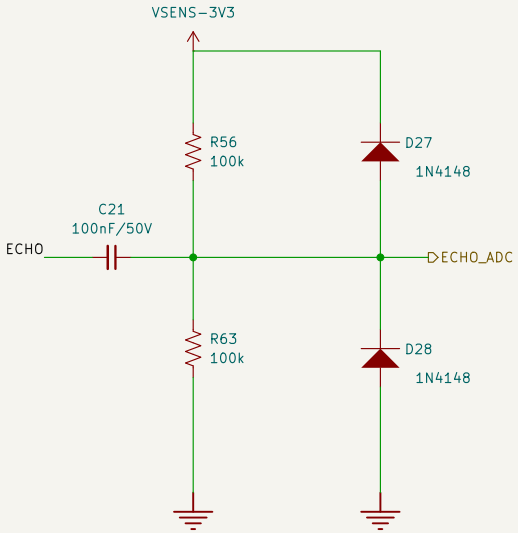
Firmware toggles WDI in software (50–1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q24 gate low through C169 and tops up C170 via R178.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, water pump, LiDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down. All parts reuse existing BOM lines.

Sheet: /WATCHDOG/	
File: WATCHDOG.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 5/16



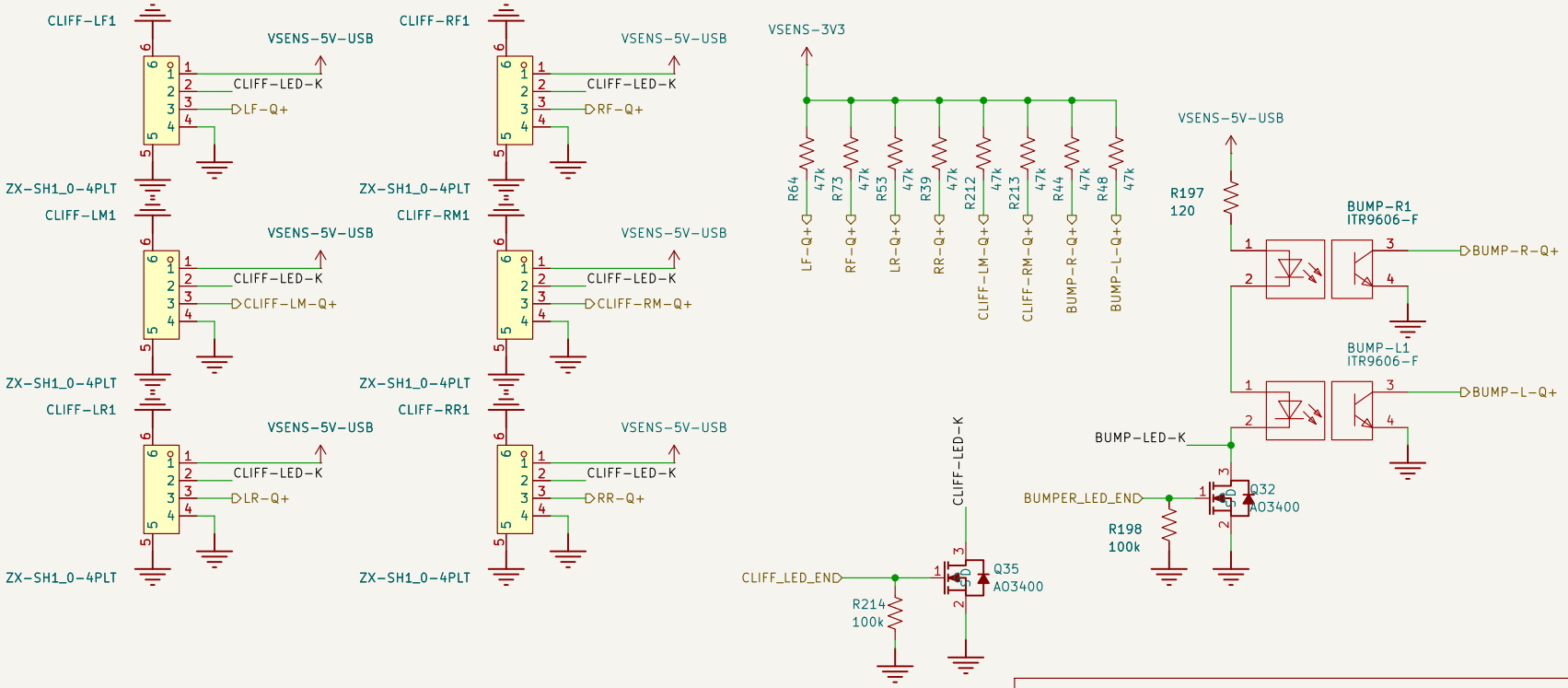
Carpet driver: one low-side AO3400 + 330R pull-up from VUS (14.4 V). The P-FET half-bridge was removed: its 100k gate pull-up could not turn it off at 290 kHz (shoot-through). R186 holds LO low in reset/flash. Drive LO in short bursts only: LO stuck high puts 0.63 W into R185.

C174 AC-couples LO to the Q18 gate: if LO sticks high, the gate decays through R186 (~10 ms) and Q18 turns off, so R185 is never stressed by firmware holding LO high.



Sheet: /CARPET-SENSOR/	
File: Carpet-sensor.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 6/16

ANTI FALL SENSORs



Cliff sensors: 6 boards (kicad/cliff-sensor), SH 4-pin vertical: 1 VSENS-5V-USB (LED, 150R on the board), 2 CLIFF-LED-K (Q35 sinks all 6 LEDs), 3 phototransistor collector (47k pull-up), 4 GND.
Modulate CLIFF_LED_EN and read each channel LED-on minus LED-off to cancel ambient light.
ADC: LF PA3, RF PA1, LR PA2, RR PA0, LM PD9, RM PE8 (LM/RM via global labels).

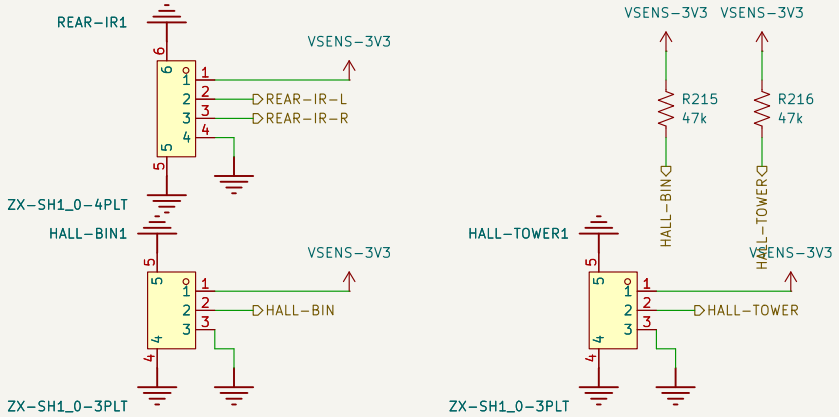
Bumpers: 2x ITR9606 slotted photo-interrupters on the main board (~11 mm tall, set by the bumper mechanism). Emitters in series from VSENS-5V-USB, switched by BUMPER_LED_EN (high = on, 100k pull-down).
BUMP-x-Q+ reads low when light passes the slot (R44/R48 47k pull-ups).

Sheet: /CLIFF/	
File: CLIFF.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 19/16



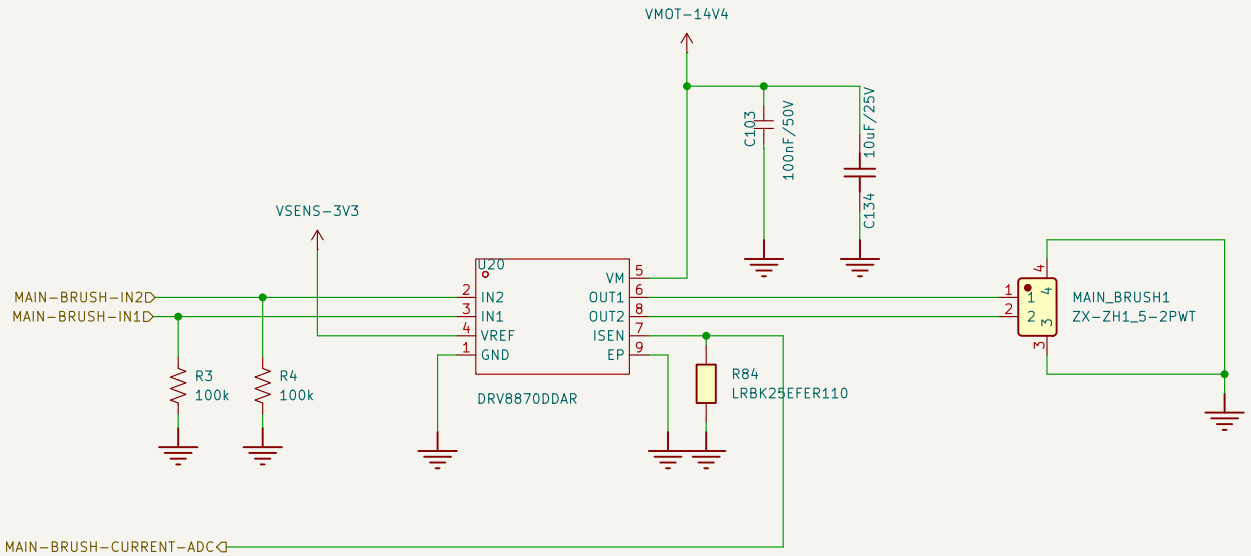
SIDE PROXI IR SENSORs

REAR DOCKING IR + HALL SENSORS

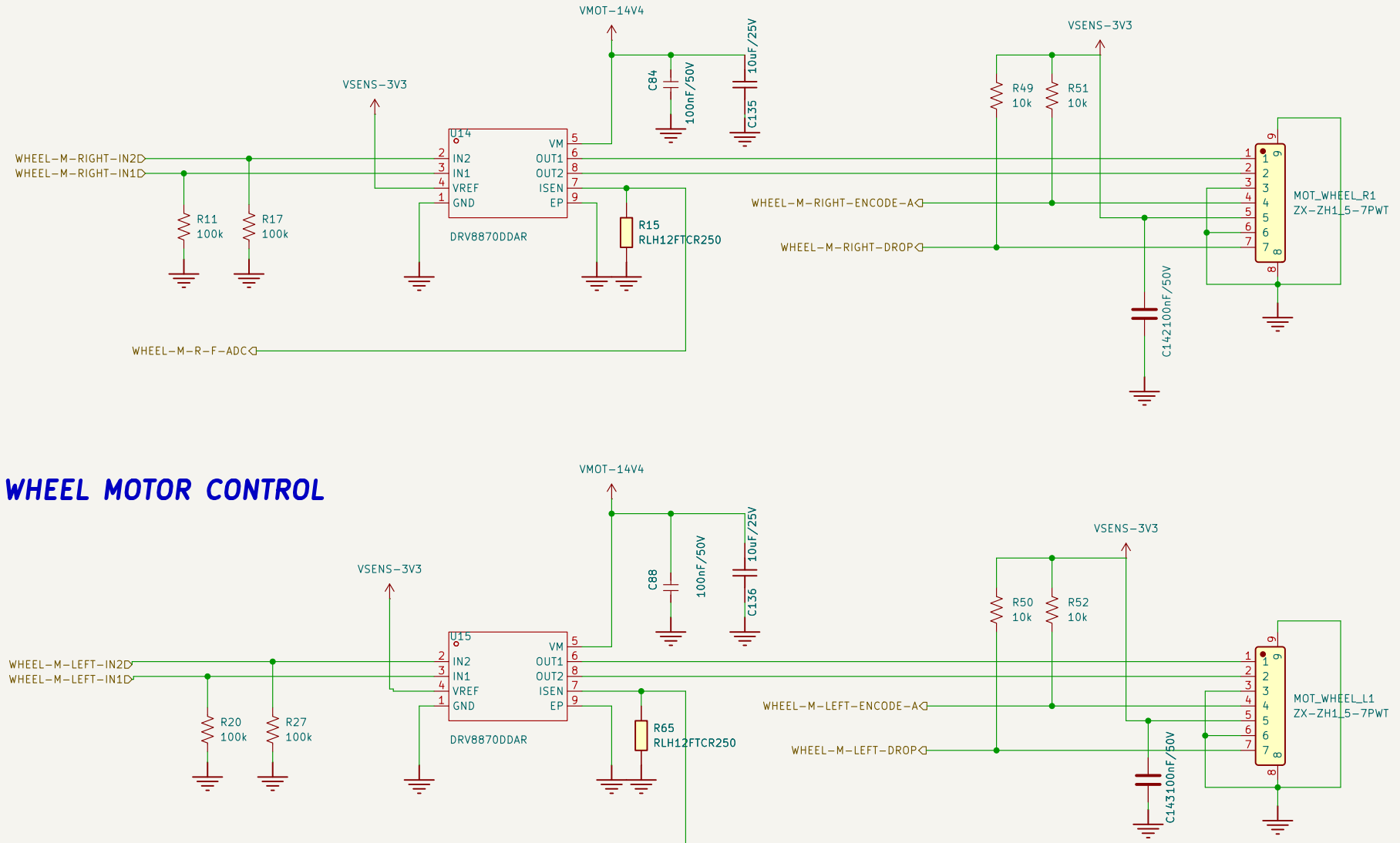


REAR-IR1: rear docking IR board, 1 3.3 V, 2 IR_L (PC8, TIM3_CH3), 3 IR_R (PC9, TIM3_CH4), 4 GND.
HALL-BIN1 / HALL-TOWER1: Hall boards, SH 3-pin (ZX-SH1.0-3PLT, so a Hall cable cannot go into a 4-pin socket): 1 3.3 V, 2 OUT (PE4 / PE5, low = magnet near), 3 GND. All mounting pads to GND.

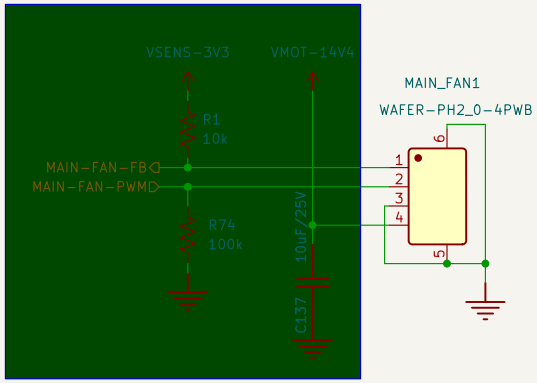
Sheet: /SIDE-PROXI/	
File: SIDE-PROXIMITY-IR-SENSOR .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 20/16



Sheet: /MAIN-BRUSH MOTOR/	
File: MAIN-BRUSH-MOTOR .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 22/16



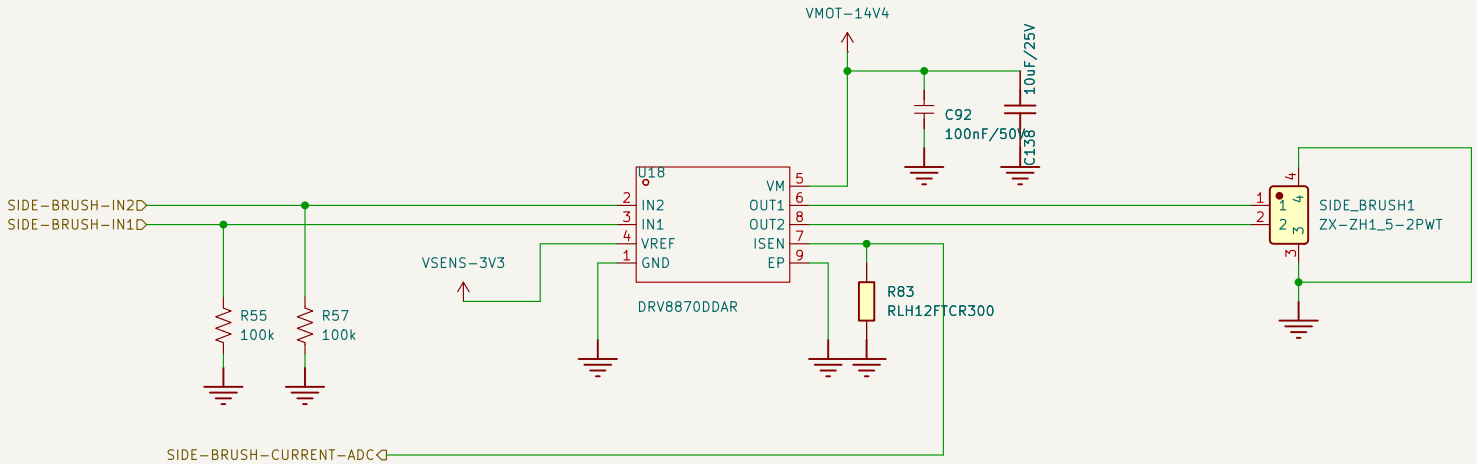
Sheet: /WHEEL MOTOR/		
File: WHEEL-MOTORs .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 23/16



- feedback signal is open collector
- set PWM LOW to turn fan OFF

Sheet: /MAIN-FAN/		
File: MAIN-FAN .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 24/16

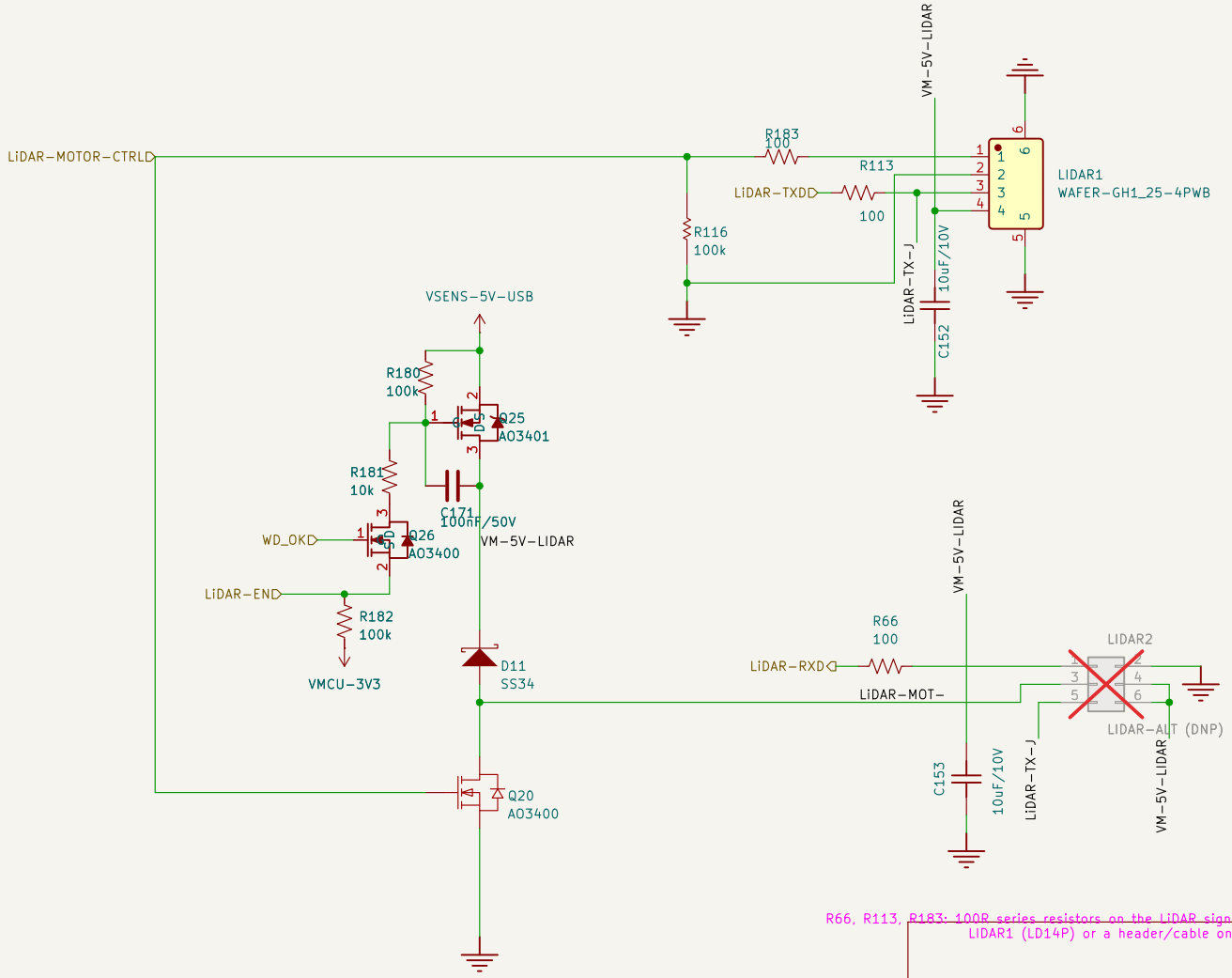
SIDE-BRUSH MOTOR CONTROL



Sheet: /SIDE-BRUSH/	
File: SIDE-BRUSH-MOTORs .kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	**Rev:** Id: 25/16

VM-5V-LIDAR powers the whole LiDAR (laser included) for any model. On only while WD_OK is high
AND LiDAR-EN is low (Q26 source-switched AND). R181/C171: ~1.5 ms soft start, ~30 ms soft stop.
LIDAR2: 2x3 0.1in through-hole pads (not fitted) for other LiDARs. Pin 6 = VM-5V-LIDAR for LiDARs
that need a separate motor supply (motor between pins 6 and 3). Pin 5 = the LiDAR TX, shared with LIDAR1 pin 3.

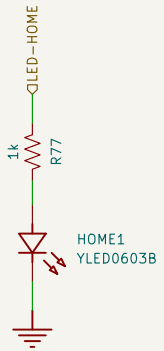
LiDAR

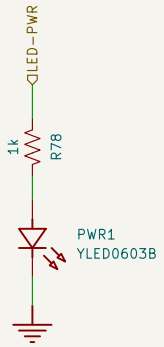


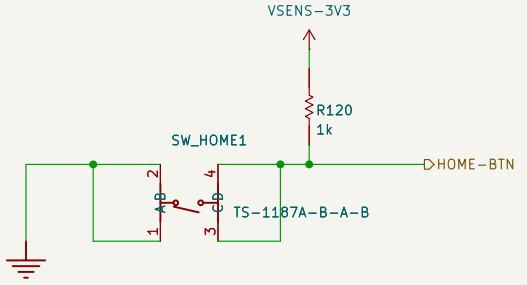
R66, R113, R183: 100R series resistors on the LiDAR signals. Only one LiDAR is fitted:
LIDAR1 (LD14P) or a header/cable on LIDAR2.

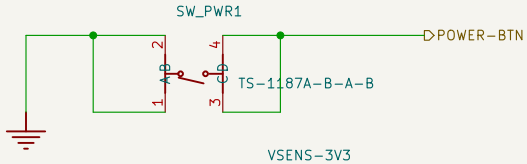
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Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 26/16

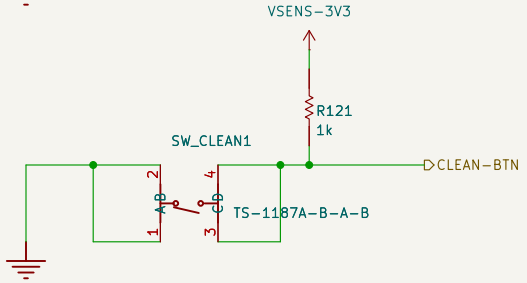
LEDs AND BUTTONs







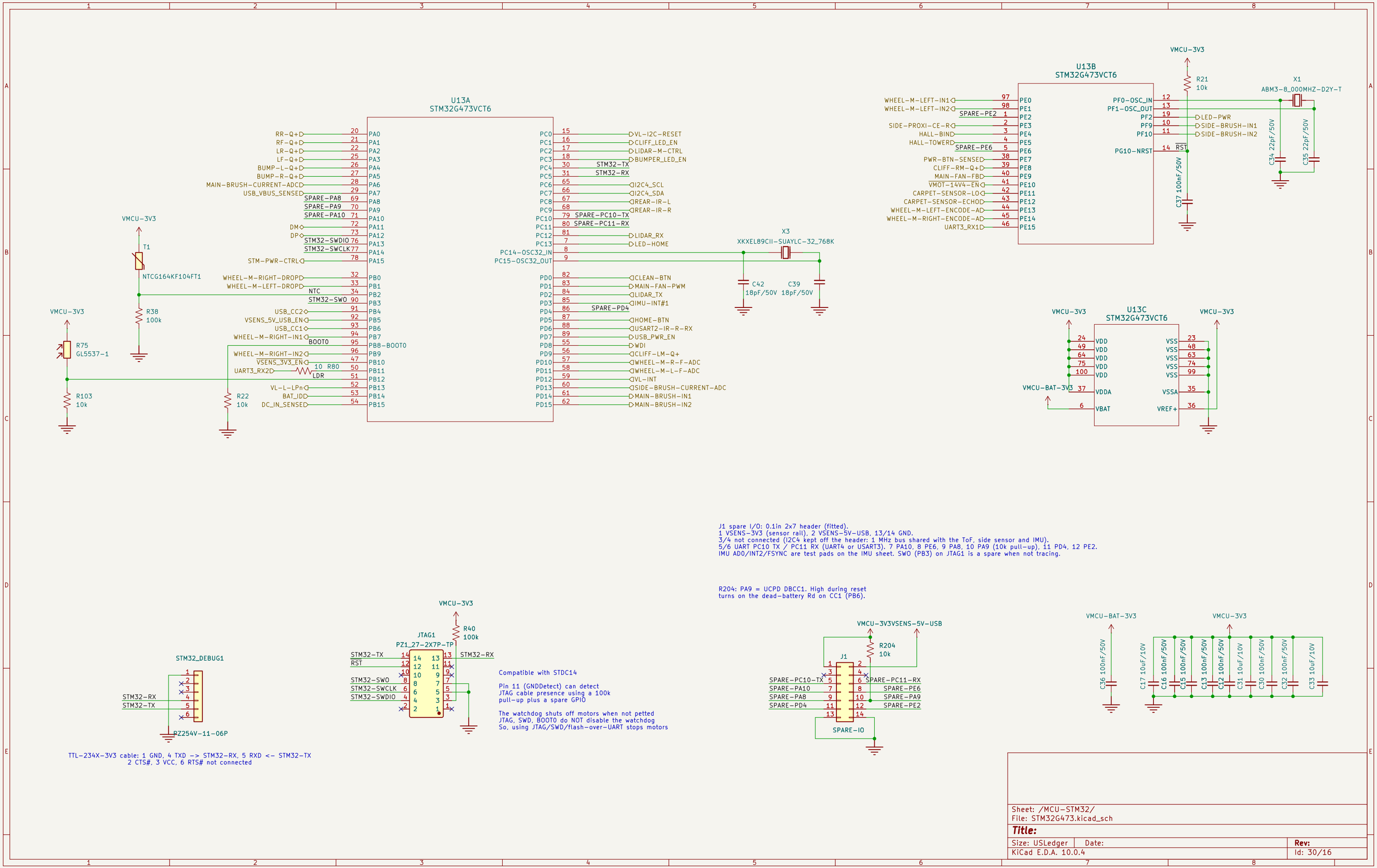




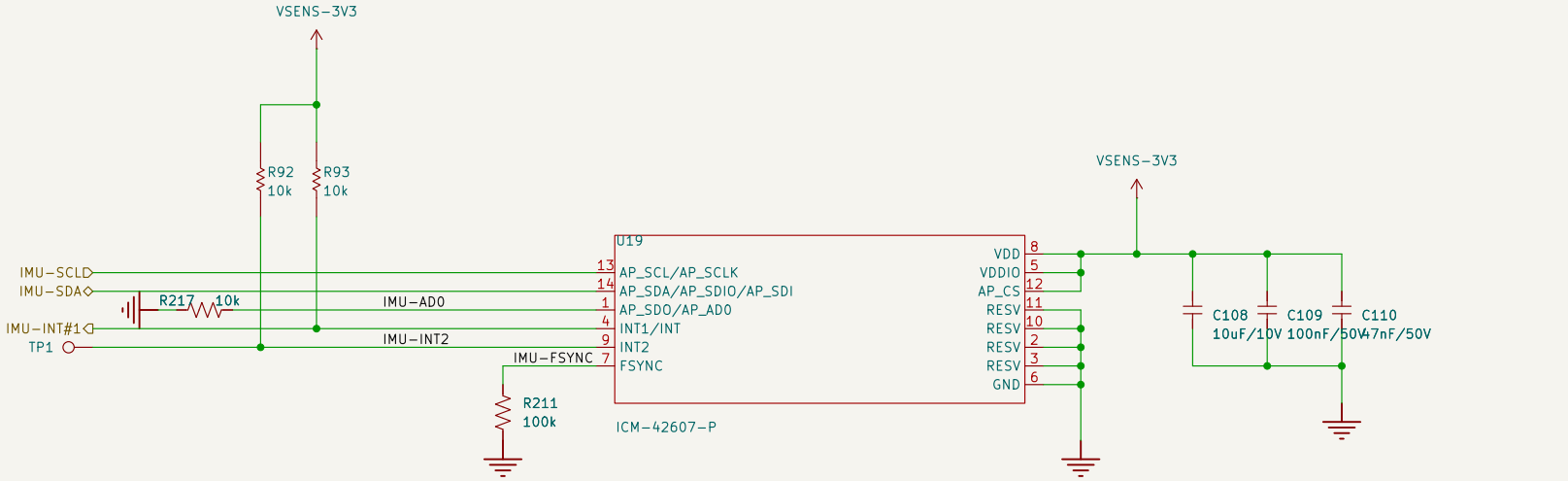
Sheet: /BUTTON-LED/
File: BUTTON-LEDs .kicad_sch

Title:

Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 28/16


U13A
STM32G473VCT6
RR-Q+ 20 PA0
RF-Q+ 21 PA1
LR-Q+ 22 PA2
LF-Q+ 25 PA3
BUMP-L-Q+ 26 PA4
BUMP-R-Q+ 27 PA5
MAIN-BRUSH-CURRENT-ADC 28 PA6
USB_VBUS_SENSED 29 PA7
SPARE-PA8 69 PA8
SPARE-PA9 70 PA9
SPARE-PA10 71 PA10
DM 72 PA11
DP 73 PA12
STM32-SWDIO 76 PA13
STM32-SWCLK 77 PA14
STM-PWR-CTRL 78 PA15
WHEEL-M-RIGHT-DROP 32 PB0
WHEEL-M-LEFT-DROP 33 PB1
NTC 34 PB2
STM32-SWO 90 PB3
USB_CC2 91 PB4
VSENS_5V_USB_EN 92 PB5
USB_CC1 93 PB6
WHEEL-M-RIGHT-IN1 95 PB7
BOOT0 PB8-BOOT0
WHEEL-M-RIGHT-IN2 96 PB9
VSENS_3V3_EN 47 PB10
10 R80
UART3_RX2 50 PB11
LDR 51 PB12
VL-L-LPn 52 PB13
BAT_IDD 53 PB14
DC_IN_SENSED 54 PB15
PC0 15 VL-I2C-RESET
PC1 16 CLIFF_LED_EN
PC2 17 LiDAR-M-CTRL
PC3 18 BUMPER_LED_EN
PC4 30 STM32-TX
PC5 31 STM32-RX
PC6 65 I2C4_SCL
PC7 66 I2C4_SDA
PC8 67 REAR-IR-L
PC9 68 REAR-IR-R
PC10 79 SPARE-PC10-TX
PC11 80 SPARE-PC11-RX
PC12 81 LIDAR_RX
PC13 7 LED-HOME
PC14-OSC32_IN 8
PC15-OSC32_OUT 9
PD0 82 CLEAN-BTN
PD1 83 MAIN-FAN-PWM
PD2 84 LIDAR_TX
PD3 85 IMU-INT#1
PD4 86 SPARE-PD4
PD5 87 HOME-BTN
PD6 88 USART2-IR-R-RX
PD7 89 USB_PWR_EN
PD8 55 WDI
PD9 56 CLIFF-LM-Q+
PD10 57 WHEEL-M-R-F-ADC
PD11 58 WHEEL-M-L-F-ADC
PD12 59 VL-INT
PD13 60 SIDE-BRUSH-CURRENT-ADC
PD14 61 MAIN-BRUSH-IN1
PD15 62 MAIN-BRUSH-IN2
X3
XKXEL89CII-SUAYLC-32_768K
C42 18pF/50V
C39 18pF/50V
VMCU-3V3
T1
NTCG164KF104FT1
R38 100k
R75 GL5537-1
R103 10k
R22 10k
U13B
STM32G473VCT6
WHEEL-M-LEFT-IN1 97 PE0
WHEEL-M-LEFT-IN2 98 PE1
SPARE-PE2 1 PE2
SIDE-PROXI-CE-R 2 PE3
HALL-BIND 3 PE4
HALL-TOWER 4 PE5
SPARE-PE6 5 PE6
PWR-BTN-SENSE 38 PE7
CLIFF-RM-Q+ 39 PE8
MAIN-FAN-FB 40 PE9
VMOT-14V4-EN 41 PE10
CARPET-SENSOR-LO 42 PE11
CARPET-SENSOR-ECHO 43 PE12
WHEEL-M-LEFT-ENCODE-AD 44 PE13
WHEEL-M-RIGHT-ENCODE-AD 45 PE14
UART3_RX1 46 PE15
PF0-OSC_IN 12
PF1-OSC_OUT 13
PF2 19 LED-PWR
PF9 10 SIDE-BRUSH-IN1
PF10 11 SIDE-BRUSH-IN2
PG10-NRST 14 RST
R21 10k
X1
ABM3-8_000MHZ-D2Y-T
C34 22pF/50V
C35 22pF/50V
C37 100nF/50V
U13C
STM32G473VCT6
VDD 24
VDD 49
VDD 64
VDD 75
VDD 100
VDDA 37
VBAT 6
VSS 23
VSS 48
VSS 63
VSS 74
VSS 99
VSSA 35
VREF+ 36
VMCU-BAT-3V3
STM32_DEBUG1
STM32-RX 4
STM32-TX 5
PZ254V-11-06P
TTL-234X-3V3 cable: 1 GND, 4 TXD -> STM32-RX, 5 RXD <- STM32-TX
2 CTS#, 3 VCC, 6 RTS# not connected
JTAG1
PZ1_27-2X7P-TP
STM32-TX 14
RST 12
STM32-SWO 8
STM32-SWCLK 6
STM32-SWDIO 4
STM32-RX 13
R40 100k
Compatible with STDC14
Pin 11 (GNDDetect) can detect
JTAG cable presence using a 100k
pull-up plus a spare GPIO
The watchdog shuts off motors when not petted
JTAG, SWD, BOOT0 do NOT disable the watchdog
So, using JTAG/SWD/flash-over-UART stops motors
J1 spare I/O: 0.1in 2x7 header (fitted).
1 VSENS-3V3 (sensor rail), 2 VSENS-5V-USB, 13/14 GND.
3/4 not connected (I2C4 kept off the header: 1 MHz bus shared with the ToF, side sensor and IMU).
5/6 UART PC10 TX / PC11 RX (UART4 or USART3). 7 PA10, 8 PE6, 9 PA8, 10 PA9 (10k pull-up), 11 PD4, 12 PE2.
IMU AD0/INT2/FSYNC are test pads on the IMU sheet. SWO (PB3) on JTAG1 is a spare when not tracing.
R204: PA9 = UCPD DBCC1. High during reset
turns on the dead-battery Rd on CC1 (PB6).
VMCU-3V3VSENS-5V-USB
R204 10k
J1
SPARE-PC10-TX 5
SPARE-PA10 7
SPARE-PA8 9
SPARE-PD4 11
SPARE-PC11-RX 6
SPARE-PE6 8
SPARE-PA9 10
SPARE-PE2 12
SPARE-IO 14
C36 100nF/50V
C17 10uF/10V
C16 100nF/50V
C15 100nF/50V
C13 100nF/50V
C12 100nF/50V
C31 10uF/10V
C30 100nF/50V
C32 100nF/50V
C33 10uF/10V
Sheet: /MCU-STM32/
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Title:
Size: USLedger
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 30/16

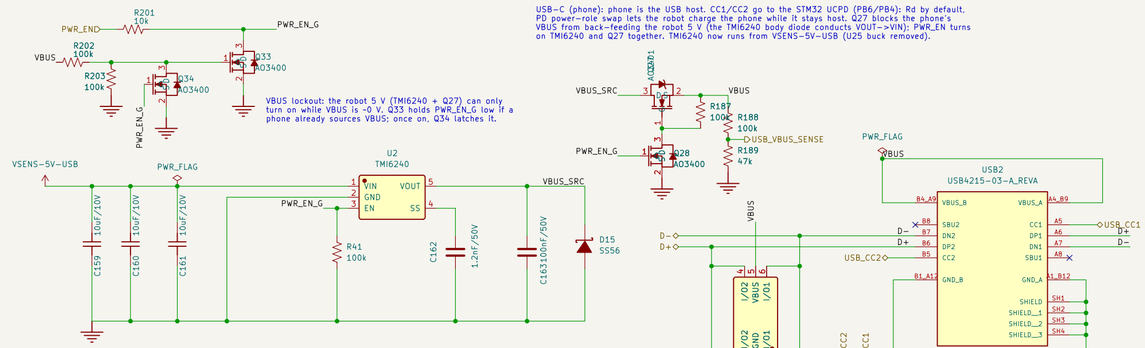
IMU-ICM-42607-P



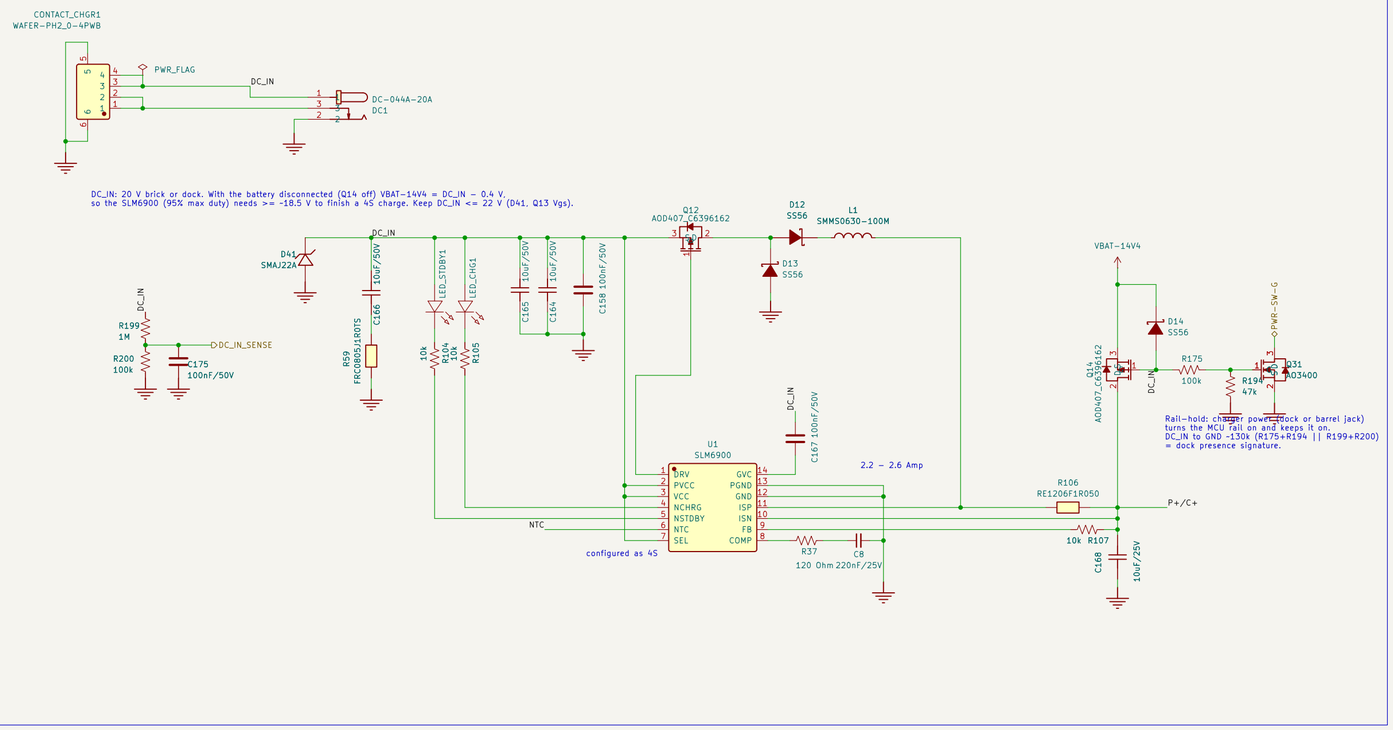
IMU on I2C4 (2.2k pull-ups on the MCU sheet). AP_AD0 pulled low by R217 (I2C address 0x68). TP1 = INT2 (test pad; the IMU is leadless). Probe AD0 on R217 and FSYNC on R211. Run a wire to a spare pin if INT2 or FSYNC is needed.

Sheet: /IMU/	
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KiCad E.D.A. 10.0.4	Id: 32/16

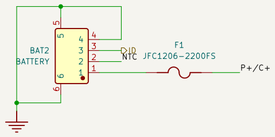
USB-C



BATTERY CHARGER



BATTERY CONNECTOR



Sheet: /BATTERY-CHARGER/	
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Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.4	**Rev:** Id: 33/16